

BILLA BALA SATYA PHANEENDRA

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EDUCATION

Koneru Lakshmaiah Education Foundation

Bachelor of Technology in Electronics and Communication Engineering

CGPA: 8.65 / 10.00

Vaddeswaram

June 2023 – Present

Rajadhani Jr Collage

SSC, Class X | %: 81 / 100

Guntur

June 2021 – March 2023

Sri Chaitanya Techno School

SSC, Class X | %: 99 / 100

Guntur

June 2020 – March 2021

SKILLS

Interests: Verilog, RTL Design, Drones, ASIC & Verification, DSP, Physical Design

Programming: C language, Java, Verilog, System Verilog, UVM

Design Software: Xilinx Vivado, Cadence Virtuoso, LTSpice, Arduino IDE, MATLAB, Synopsys Design Compiler

Skill: AI Prompt Engineering

PROJECTS

Integrated V2X and Multi-Band Beamforming Antenna for Autonomous Vehicles

April 2026 – Present

- Developed a conformal multi-band high-gain beamforming antenna system for V2X communication and K/Ka-band radar applications in autonomous vehicles
- Implemented high-gain beamforming techniques to enhance vehicular communication, radar sensing, and intelligent transportation system performance

URAT Synthesis using Vivado

February 2026

- I have analyzed post-synthesis reports, used checkpoints for iterative analysis, and observed the impact of different synthesis settings on performance.

CMOS S-Parameter based Low Noise Amplifier

OCT 2025 – NOV 2025

- Designed and simulated a 2.4 GHz CMOS Low Noise Amplifier using S-parameter analysis, achieving a gain below 15 dB.

Autonomous Quadcopter Drone

Jan 2025 – April 2025

- Implemented autonomous features like real-time monitoring for performance analysis and optimization
- Presented the project in Electronic System Design Skill Development Course Review (May 2025)

Home Automation System using the STM32

September 2024

- Executed a Bluetooth-controlled LED automation prototype using the STM32 Blue Pill and HC-05 module, enabling wireless ON/OFF Control of three LEDs through a mobile application.

Detection Based Security System

May 2024 – July 2024

- Designed and implemented a real-time Security System using an ESP8266 Node MCU.

ACHIEVEMENTS

IEEE AP-S Undergraduate Research Funding Award – 2026

- Awarded a USD 3,000 research grant by the IEEE Antennas and Propagation Society (IEEE AP-S) under the IEEE AP-S Undergraduate Research Funding Program 2026.
- Selected for the project: "A Conformal Integrated-Aperture Multi-Band Antenna System for Intelligent Vehicles: V2X (5.9 GHz) and High-Gain Beamforming at K/Ka-Bands (24/28/39 GHz) for Radar-Communication in Autonomous Vehicles."

RESEARCH PUBLICATION

Artificial Intelligence-powered Drone for Wildlife Monitoring and Animal Detection

Accepted for presentation at WAMS 2026 Conference (Paper ID: 17)

INTERSHIPS

AICTE Virtual Internship by Microchip

04/2025 – 06/2025

Embedded System Developer

Skill Dzure

05/2025 – 06/2025

VLSI Internship Program

HACKATHONS

SDG Innovation Challenge -Techno Summit'25

September 2025

- Sathyabama University
- Secured 4th place at the SDG Innovation Challenge.
- Developed NP-Scan for Smart Pesticide Screening aligned with SDG3. Research Existing detection methods and key challenges.

CERTIFICATION

Cambridge Linguaskill Test (Achieved B1 level)

July 2024

Cambridge

Artificial Intelligence and Machine Learning Using Python

September 2024

Taras Systems and Solutions

Altium Designer Essentials: Mastering the Fundamental

January 2025

Altium

Embedded System Application and IOT Programming

April 2025

Taras Systems and Solutions

Spectre Simulator Fundamentals S1

April 2025

Cadence
VLSI Design and Verification – L1
Taras Systems and Solutions
System Verilog & UVM -L2
Taras Systems and Solutions

December 2025

March 2026

POSITIONS OF RESPONSIBILITY

Vice President, PULSE ECE Student Body KL University	April 2026 – Present
Chair, IEEE Antennas and Propagation Society (AP'S) Student Branch KL University	January 2026 – Present
KL University Alumni Relations Chief Student Coordinator	April 2026 – Present
NSS Core Unit-8 KLEF Campus relations	January 2026 – present
KL researcher's Club Financial Lead	September 2024 – October 2025

EXTRA CIRCULARS ACTIVITIES

Organized Samyak -25 • Headed Sponsorship and Partnership operations as Chief. • Coordinated with marketing and event sponsor visibility.	October 2025
Organized Surabhi - 26 • Organized guest accommodation, transportation as Lead Hospitality • Resolved on-ground issues efficiently, improving guest experience.	March 2026